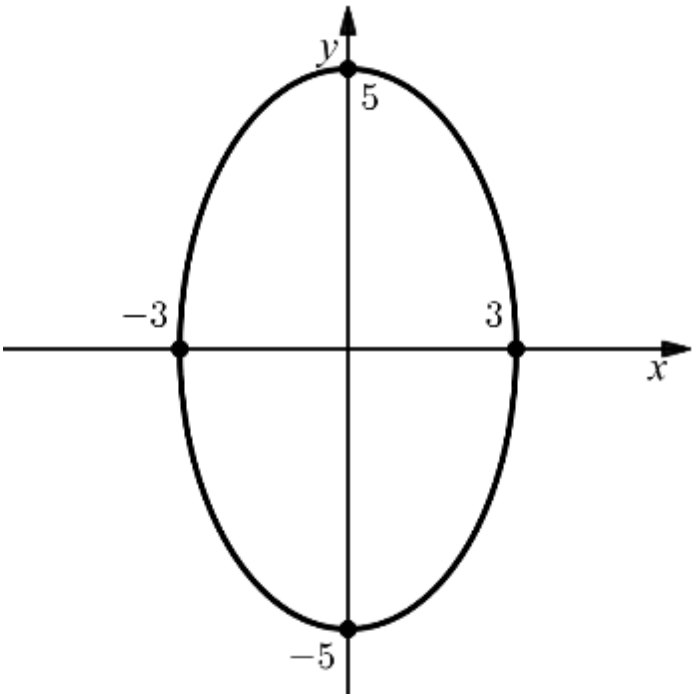
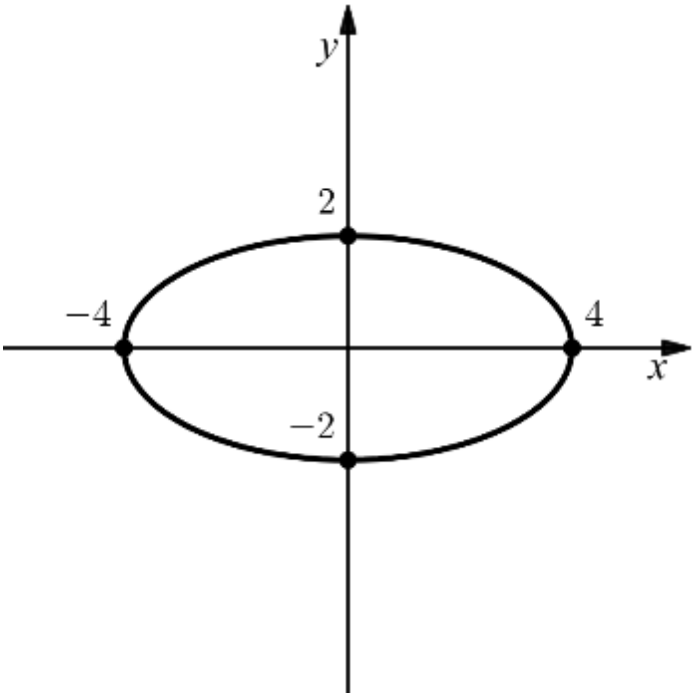
1.22 Graph Sketching: Ellipses

Worked Solutions

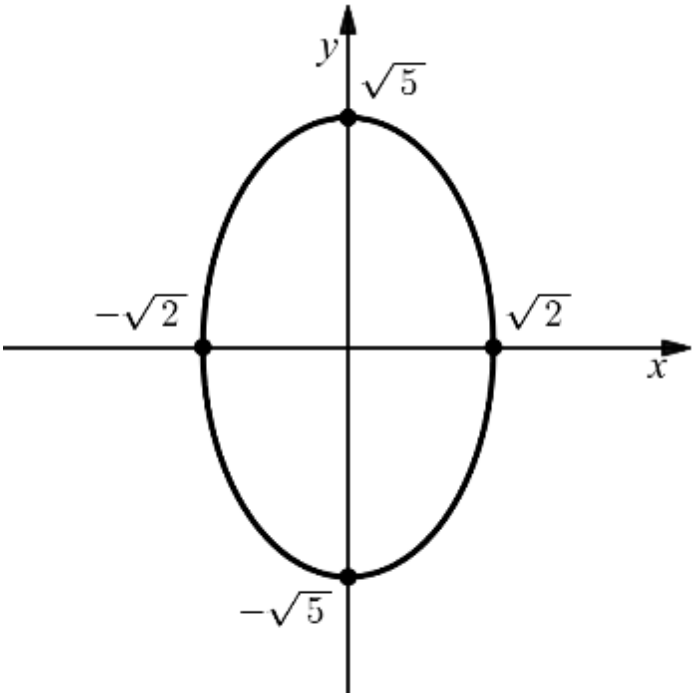
1 a)



1 b)

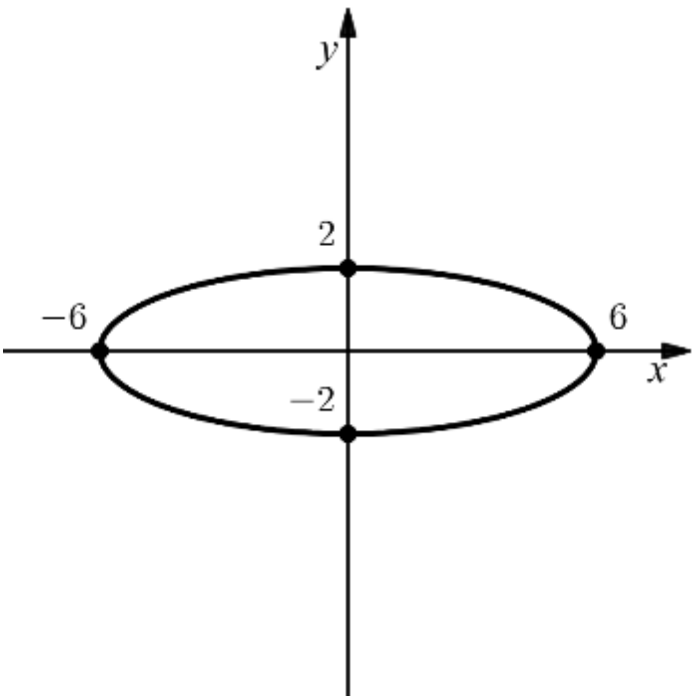


1 c)



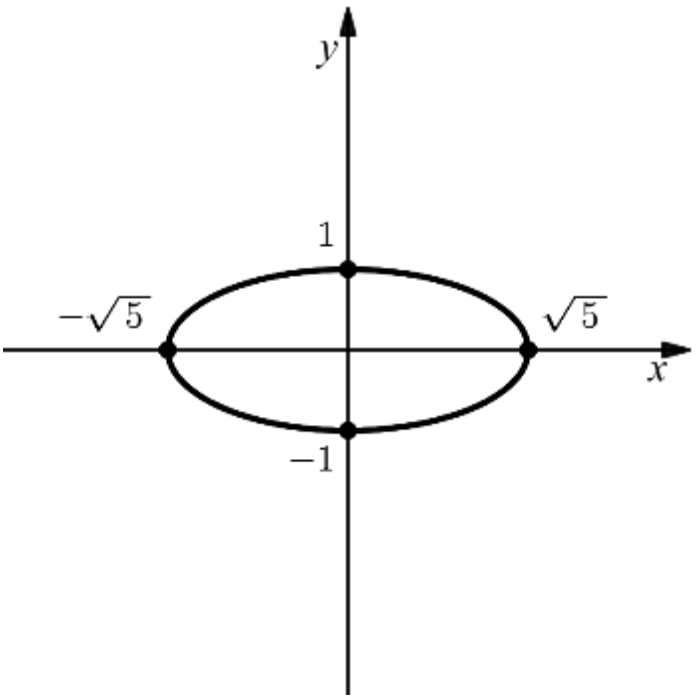
1 d)

$\frac{x^2}{36} + \frac{y^2}{4} = 1$

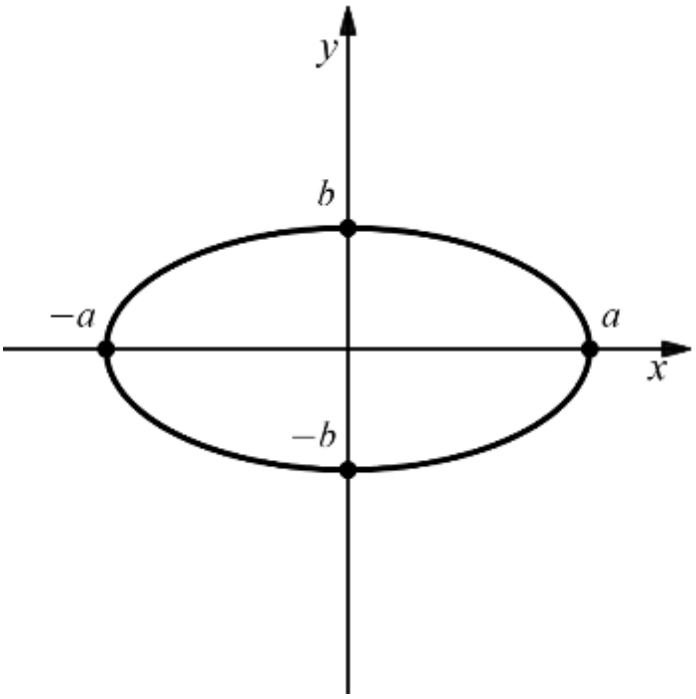


1 e)

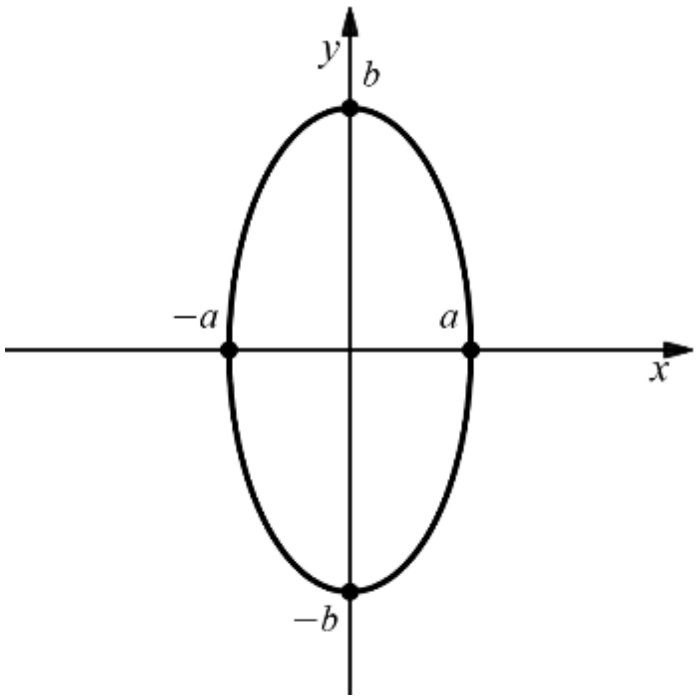
$\frac{x^2}{5} + y^2 = 1$



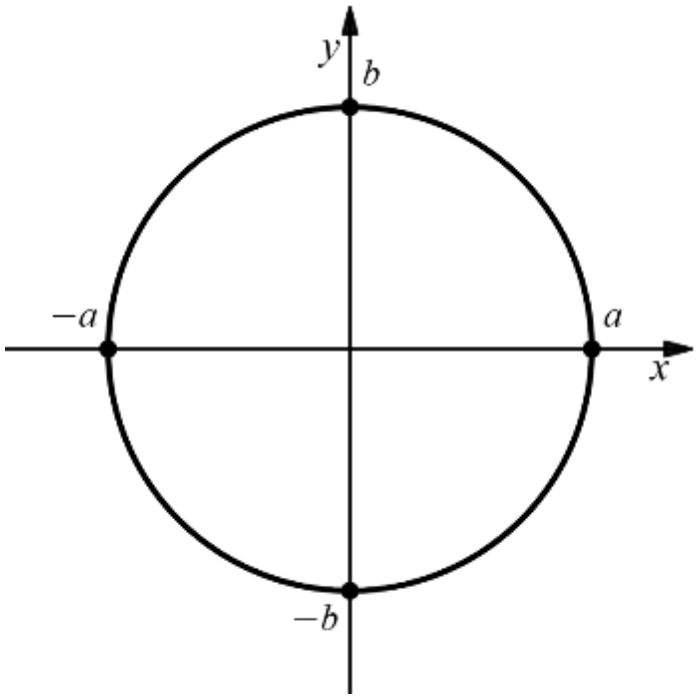
1 f)



1 g)



1 h)



2 a)

$\frac{x^2}{5} + \frac{y^2}{3} = 1$ so $(\sqrt{5}, 0)$ and $(-\sqrt{5}, 0)$

So $x = -\sqrt{5}$ and $x = \sqrt{5}$

2 b)

$\frac{x^2}{3} + \frac{y^2}{5} = 1$ so $(\sqrt{3}, 0)$ and $(-\sqrt{3}, 0)$

So $x = -\sqrt{3}$ and $x = \sqrt{3}$

3 a)

$(2, 0)$ gives $a = 2$

$(0, 4)$ gives $b = 4$

$a = 2$ and $b = 4$

3 b)

$(3, 0)$ gives $a = 3$

$(0, 2)$ gives $b = 2$

$a = 3$ and $b = 2$

3 c)

$(\sqrt{7}, 0)$ gives $a = \sqrt{7}$

$(0, \sqrt{3})$ gives $b = \sqrt{3}$

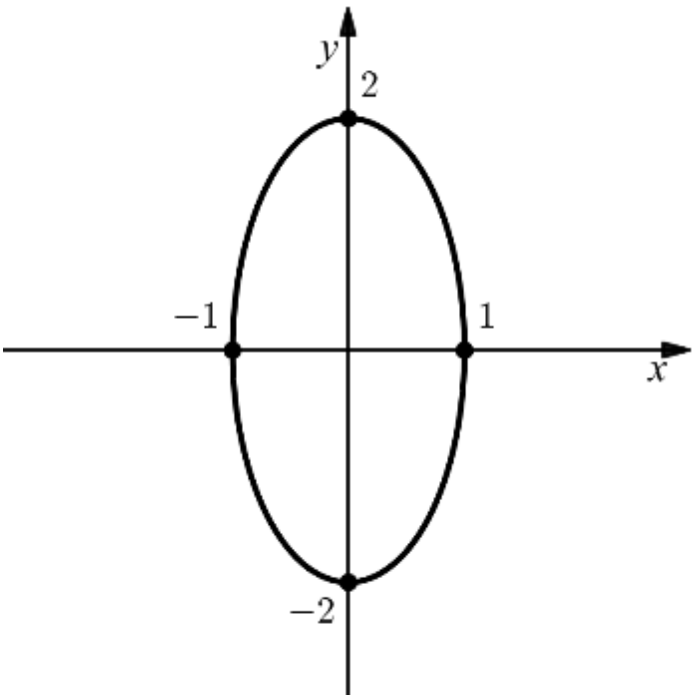
$a = \sqrt{7}$ and $b = \sqrt{3}$

4 a)

$\frac{1}{a} = 1$ and $\frac{4}{b} = 1$

$\therefore a = 1$ and $b = 4$

4 b)

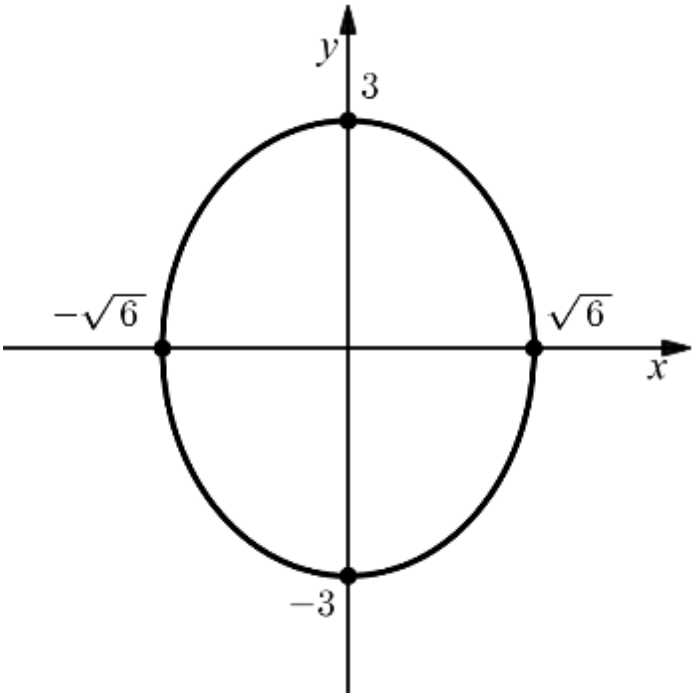


5 a)

$\frac{6}{a} = 1$ and $\frac{9}{b} = 1$

$\therefore a = 6$ and $b = 9$

5 b)



6 a)

$$\frac{4}{a} + \frac{6}{b} = 1$$

$$\frac{6}{a} + \frac{3}{b} = 1 \Rightarrow \frac{12}{a} + \frac{6}{b} = 2$$

which gives

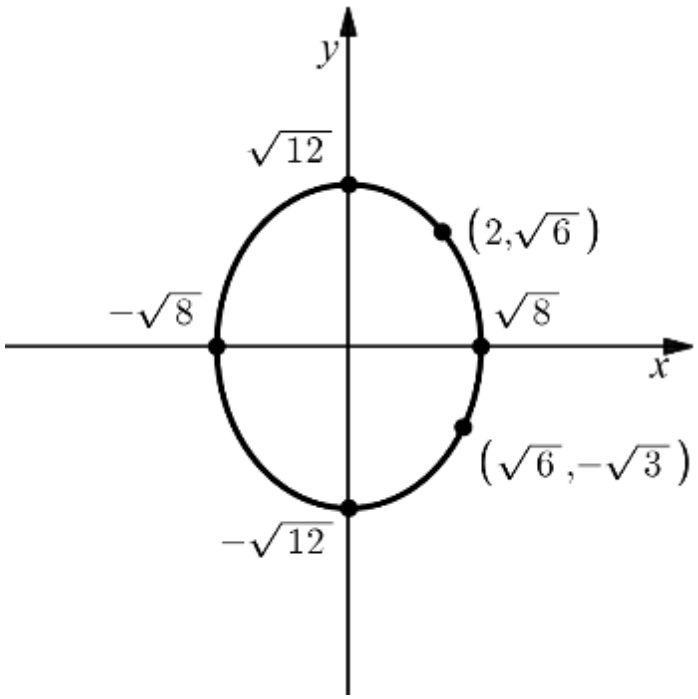
$$\frac{4}{a} + (2 - \frac{12}{a}) = 1$$

$1 = \frac{8}{a}$ so $a = 8$

$$\frac{4}{8} + \frac{6}{b} = 1 \Rightarrow \frac{b}{6} = 2$$

$\therefore a = 8$ and $b = 12$

6 b)



7 a)

When $y = 0$, $x^2 = a$

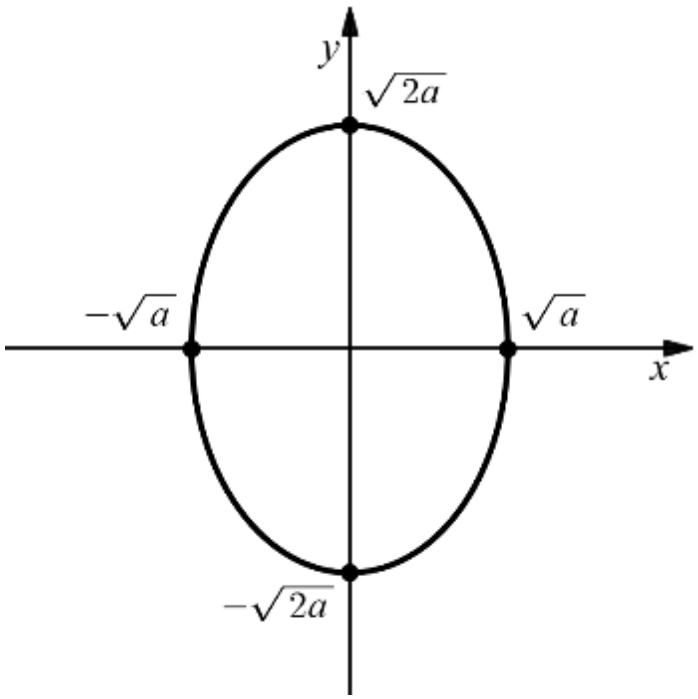
$$\therefore (-\sqrt{a}, 0) \text{ and } (\sqrt{a}, 0)$$

7 b)

When $x = 0$, $y^2 = 2a$

$$\therefore (0, -\sqrt{2a}) \text{ and } (0, \sqrt{2a})$$

7 c)



7 d)

$\frac{4}{a} + \frac{10}{2a} = 1 \Rightarrow \frac{9}{a} = 1$

$$\therefore a = 9$$